

---

## Symmetries in Physics — Exercise Sheet 13

### *The Heisenberg group and Fock space*

#### Variant: Questions only

---

**Exercise 1** Realise the Heisenberg group by upper unitriangular matrices  $g(a, b, c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$  and the algebra by  $X(a, b, c) = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}$ . (i) Show  $e^{X(a,b,c)} = g(a, b, c + \frac{1}{2}ab)$ . (ii) Verify the exact Baker–Campbell–Hausdorff law by computing both  $g(a, b, c)g(a', b', c')$  and the bracket series, and confirm they agree.

**Exercise 2** The Weyl form of the canonical commutation relations uses the unitaries  $U(a) = e^{-iaP/\hbar}$  and  $V(b) = e^{ibQ/\hbar}$  on  $L^2(\mathbb{R})$  — translations and modulations — which satisfy  $U(a)V(b) = e^{-iab/\hbar}V(b)U(a)$ . Show that the Schrödinger representation is irreducible: any bounded  $T$  commuting with all  $U(a), V(b)$  is a scalar. You may use these point-of-use imports: the von Neumann algebra generated by the  $V(b)$  is  $\mathcal{M} = \{M_f : f \in L^\infty(\mathbb{R})\}$ , and on the sigma-finite space  $L^2(\mathbb{R})$  this multiplication algebra is maximal abelian,  $\mathcal{M}' = \mathcal{M}$ . In the last step, treat translation invariance as equality of  $L^\infty$  classes, not as a pointwise identity.

**Exercise 3** Coherent states are  $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ , the eigenvectors  $a|\alpha\rangle = \alpha|\alpha\rangle$ . (i) Compute  $\Delta Q^2$  and  $\Delta P^2$  and show the product is the minimum  $\hbar^2/4$ . (ii) Prove the resolution of identity  $\frac{1}{\pi} \int |\alpha\rangle \langle \alpha| d^2\alpha = \mathbb{K}$ . (iii) Under  $H = \hbar\omega(a^\dagger a + \frac{1}{2})$ , show a coherent state stays coherent with  $\alpha(t) = \alpha e^{-i\omega t}$ , and deduce that  $\langle Q \rangle(t), \langle P \rangle(t)$  trace the classical oscillator ellipse. For part (ii), interpret the integral weakly: first test its matrix elements between vectors in  $\mathcal{D}_{\text{fin}} = \text{span}\{|n\rangle : n \geq 0\}$ . You may use the elementary extension lemma: a bounded sesquilinear form on a dense subspace has a unique bounded extension to the Hilbert space.

**Exercise 4** Build the fermionic Fock representation for  $d = 2$  explicitly: operators  $c_1, c_2$  with  $\{c_i, c_j\} = 0$ ,  $\{c_i, c_j^\dagger\} = \delta_{ij}$ , as  $4 \times 4$  matrices via a Jordan–Wigner construction.

**Exercise 5** For the bosonic Fock space over  $\mathbb{C}^d$ , let  $\mathcal{F}_{\text{alg}} = \bigoplus_{n \geq 0}^{\text{alg}} \text{Sym}^n \mathbb{C}^d$  and define on this core  $d\Gamma(A) = \sum_{ij} A_{ij} a_i^\dagger a_j$  for a  $d \times d$  matrix  $A$ . (i) Show  $[d\Gamma(A), a_k^\dagger] = a_{Ae_k}^\dagger$  and deduce that  $d\Gamma$  is a Lie-algebra homomorphism on this common invariant core. (ii) Identify the number operator as  $d\Gamma(\mathbb{K})$  and show it is unbounded. For  $U \in \text{U}(d)$  define the all-Fock unitary  $\Gamma(U) = \bigoplus_n U^{\otimes n}$  on the symmetric sectors and prove  $\Gamma(U) a_i^\dagger \Gamma(U)^{-1} = a_{Ue_i}^\dagger$ . If  $A$  is anti-Hermitian, differentiate at  $t = 0$  on  $\mathcal{F}_{\text{alg}}$  and state the imported closure formula relating  $\Gamma(e^{tA})$  to  $d\Gamma(A)$ .

**Exercise 6** Two identical oscillators coupled by  $H = \frac{p_1^2 + p_2^2}{2m} + \frac{1}{2}m\omega^2(x_1^2 + x_2^2) + \frac{1}{2}m\kappa(x_1 - x_2)^2$ . Diagonalise by a normal-mode change of variables, give the spectrum, and identify the role of the exchange symmetry  $S_2$ .